Solar O appendix

Sun · O · downloadable appendix

Document	Solar O appendix
Subject	Solar photospheric abundance appendix · O
Author	Ryan Schmitt
Date	2026-10-29
Publisher	The Exoplanet Codex · exoplanetcodex.org
Version	feed v1.102
Source	science commit 46fd923d5101
Web	https://exoplanetcodex.org/systems/sol/elements/o/

Solar oxygen: 8.757 ± 0.054

Asplund, Amarsi & Grevesse 2021: 8.69 ± 0.04 · this measurement +0.067 dex

Combined the way Asplund, Amarsi & Grevesse (2021) combine it: weighted mean and error of the indicator families [O I], O I, OH (Asplund+2021 Sect. 2.3); 1D results moved to the 3D scale by Asplund+2021 Table 3 (3D minus 1D MARCS for the same indicator). Each indicator is represented by our best analysis of it.

[O I]: 8.973 ± 0.312 (1D 8.934 +0.039 to 3D) · HARPS · VIS · [O I] 6300 Å forbidden line (blended with Ni I) · 1D LTE

O I: 8.751 ± 0.054 · Kitt Peak (Kurucz 2005) · red-optical · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi)

Not measurable in our spectra: OH.

Highlighted Products

VIS

8.934 ± 0.312

[O I] 6300 Å forbidden line (blended with Ni I) · 1D LTE · HARPS · Codex Grade

red-optical

8.751 ± 0.054

O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (Kurucz 2005) · Reference Grade

NIR

8.817 ± 0.055

O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · Kitt Peak (Kurucz 2005) · Reference Grade

Line grades

Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	18 products -- O I · Asplund et al. 2021 lines
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	10 products -- O I · Codex-graded lines; [O I] 6300 Å forbidden line (blended with Ni I)

Grade	What it means	This element
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

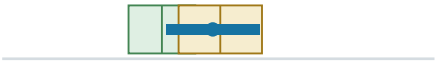
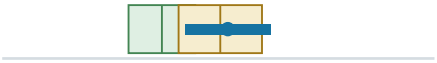
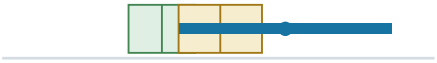
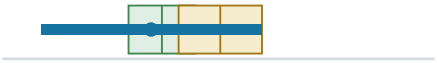
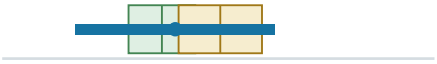
Error-bar Forest Plot

VIS — HARPS
A(O) dex; axis 8.59–9.28

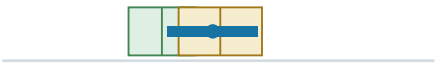
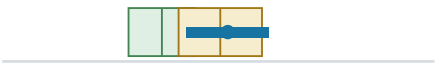
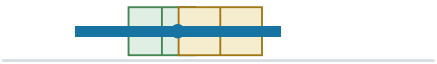
[O I] 6300 Å forbidden line (blended with Ni I) · 1D LTE Codex Grade · 1 line		8.934 ±0.312

VIS axis: 8.59 · 8.76 · 8.93 · 9.11 · 9.28 A(O) dex

red-optical — IAG FTS
A(O) dex; axis 8.50–9.02

O I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 4 lines		8.922 ±0.058
O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 4 lines		8.751 ±0.056
O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 4 lines		8.769 ±0.052
O I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		8.838 ±0.127
O I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		8.677 ±0.132
O I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines		8.706 ±0.120

red-optical — Kitt Peak (Kurucz 2005)
A(O) dex; axis 8.50–9.02

O I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 4 lines		8.922 ±0.056
O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 4 lines		8.751 ±0.054
O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 4 lines		8.769 ±0.049
O I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		8.841 ±0.131
O I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines		8.680 ±0.136
O I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines		8.709 ±0.123

red-optical — Kitt Peak (1984, molecfit)
A(O) dex; axis 8.50–9.02

O I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 4 lines	8.925 ±0.062
O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 4 lines	8.754 ±0.060
O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 4 lines	8.771 ±0.054
O I · Codex-graded lines · 1D LTE Codex Grade · 2 lines	8.833 ±0.140
O I · Codex-graded lines · 3D NLTE (Amarsi) Codex Grade · 2 lines	8.672 ±0.143
O I · Codex-graded lines · 1D NLTE (Bergemann) Codex Grade · 2 lines	8.701 ±0.130

red-optical axis: 8.50 · 8.63 · 8.76 · 8.89 · 9.02 A(O) dex

NIR — IAG FTS
A(O) dex; axis 8.53–8.99

O I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 3 lines	8.742 ±0.085
O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines	8.651 ±0.087
O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 3 lines	8.689 ±0.082

NIR — Kitt Peak (Kurucz 2005)
A(O) dex; axis 8.53–8.99

O I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 3 lines	8.910 ±0.053
O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines	8.817 ±0.055
O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) Reference Grade · 3 lines	8.848 ±0.051

NIR — Kitt Peak (1984, molecfit)
A(O) dex; axis 8.53–8.99

O I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 3 lines	8.825 ±0.064
O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) Reference Grade · 3 lines	8.733 ±0.065

NIR — Kitt Peak (1984, molecfit)
A(O) dex; axis 8.53–8.99

O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann)
Reference Grade · 3 lines



8.768
±0.061

NIR axis: 8.53 · 8.65 · 8.76 · 8.88 · 8.99 A(O) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). For a molecule the NLTE term is replaced by its dependence on the other element, and the 3D term comes from the 3D-to-1D difference Amarsi et al. 2021 publish for the same band. Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every oxygen value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the oxygen abundance as the free parameter. The synthesis includes the atomic lines and the molecular bands (CN, CH, C_2, OH and others) in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, blends and coupling to the other C/N/O abundances, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	VIS
What was measured	[O I] 6300 Å forbidden line (blended with Ni I)
Solar spectra	HARPS
Treatments	1D-LTE
Lines	1
Result A(O)	8.934 (1D-LTE)
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.31, continuum placement 0.17

Band	red-optical
What was measured	O I · Codex-graded lines
Solar spectra	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, 3D-NLTE
Lines	2
Result A(O)	8.672–8.680 (3D-NLTE)

Band	red-optical
What was measured	O I · Codex-graded lines
Largest uncertainty terms (dex)	fit / line-to-line scatter 0.11, NLTE / 3D correction 0.07

Band	red-optical
What was measured	O I · Asplund et al. 2021 lines
Solar spectra	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, 3D-NLTE
Lines	4
Result A(O)	8.751–8.754 (3D-NLTE)
Largest uncertainty terms (dex)	continuum placement 0.05, model atmosphere 0.03

Band	NIR
What was measured	O I · Asplund et al. 2021 lines
Solar spectra	IAG FTS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE, 3D-NLTE
Lines	3
Result A(O)	8.651–8.817 (3D-NLTE)
Largest uncertainty terms (dex)	continuum placement 0.14, model atmosphere 0.03

Solar oxygen context

Oxygen is traced by forbidden [O I], permitted O I and molecular OH indicators where available. Blends, departures from LTE and atmospheric structure affect these families differently, so their results retain separate identities.

C/O indicator strategy · Methodology

Open provenance decisions

O I 615.8nm: DISAGREES — reported to Ryan, not resolved. . Store log gf -0.409 (); summed pull -0.2966 (B+).

O I 630.0nm: DISAGREES — reported to Ryan, not resolved. store claims A, pull publishes B+. Store log gf -9.717 (A); summed pull -9.7744 (B+).

OPEN. Tachiev and NIST differ systematically by -0.016 dex on the 777 triplet and -0.005 dex on 8446. AGSS21 names Tachiev for N i only and states no O i gf source. Both values are reported in oi_tachiev_vs_nist.csv; NIST is what the store already carried and this script does not change it. Ryan decides.

Adjudication record · Indicator gf checks

Download the evidence

Download PDF — Solar O appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit 46fd923d510159bad5c0aaf5eeb616e175e048e5
Feed O.json v1.102 · updated 2026-10-10T15:46:17Z
Generated 2026-10-29T07:34:56-06:00 (source commit timestamp)
Feed SHA-256 1cb9f530ec8ef1256b0ea62505c70d49cf4960b3bd9542e0234cec10b48150f8
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar O atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [1], [2].

Atomic data and gf authorities. The recorded O I/O II base pedigree is Wiese–Fuhr–Deters / Opacity Project theory. Per-transition exceptions and the open O I source decisions remain explicit. Sources: [3], [4], [5].

Atomic data and gf authorities. O I triplet source record: NIST — Tachiev NOT adopted; OPEN — under review Sources: [3], [6].

Model and grid sources. Registered O I line/grid lineage: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — line-by-line 3D-NLTE grids built on the Amarsi+2018 O I model atom. The current feed treatment remains authoritative; a grid reference does not imply a correction was applied. Sources: [7], [8].

Atomic data and gf authorities. Atomic O I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [3], [4], [5].

Model and grid sources. Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [9].

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 3D non-LTE - 1D LTE on STAGGER (table2 C I / table3 O I); PER-LINE additive correction, grid lines only (match <= 0.05 A) Sources: [7], [8].

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 1D non-LTE - 1D LTE (table5 C I / table6 O I); PER-LINE additive correction, grid lines only (match <= 0.05 A) Sources: [7], [8].

Special diagnostics. The synthesis sidecar names Johansson’s Ni I blend data, Storey & Zeippen’s forbidden-line gf lineage and Caffau’s CO5BOLD comparison anchor. The feed entry remains 1D-LTE and quarantined; the sidecar’s cited anchor is not promoted to a measured abundance. Sources: [10], [11], [12].

Product-to-source index

NIR · IAG FTS · O I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [5], [9] O-products-2f26fcc4de61

NIR · IAG FTS · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-ee093b5cd43b

NIR · IAG FTS · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-dcccc1bf145a

NIR · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [5], [9] O-products-7570ea595567

NIR · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-34b4e514053b

NIR · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-f76024e80aca

NIR · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [5], [9] O-products-97dff9c24866

NIR · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-4ce6582718b9

NIR · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-dbd116adb0fe

VIS · HARPS · [O I] 6300 Å forbidden line (blended with Ni I) · 1D LTE · [1], [3], [4], [5], [9], [10], [11], [12] O-products-951619c74bb3

red-optical · IAG FTS · O I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [5], [9] O-products-400d2fb23109

red-optical · IAG FTS · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-b43c168af108

red-optical · IAG FTS · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-14585017d015

red-optical · IAG FTS · O I · Codex-graded lines · 1D LTE · [1], [3], [4], [5], [9] O-products-4b0ab56d8136

red-optical · IAG FTS · O I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-b35970fb8af7

red-optical · IAG FTS · O I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-9f2918c00938

red-optical · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [5], [9] O-products-b87159d4fc51

red-optical · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-1a0fed49853d

red-optical · Kitt Peak (Kurucz 2005) · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-30cc32d77032

red-optical · Kitt Peak (Kurucz 2005) · O I · Codex-graded lines · 1D LTE · [1], [3], [4], [5], [9] O-products-5dd7a576c91e

red-optical · Kitt Peak (Kurucz 2005) · O I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-fabf2e672c03

red-optical · Kitt Peak (Kurucz 2005) · O I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-b90f07a8459c

red-optical · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 1D LTE · [1], [3], [4], [5], [9] O-products-192c0fab28e1

red-optical · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-882b5f8ac19f

red-optical · Kitt Peak (1984, molecfit) · O I · Asplund et al. 2021 lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-6fccf43002bd

red-optical · Kitt Peak (1984, molecfit) · O I · Codex-graded lines · 1D LTE · [1], [3], [4], [5], [9] O-products-e1b890ab2006

red-optical · Kitt Peak (1984, molecfit) · O I · Codex-graded lines · 3D NLTE (Amarsi) · [1], [3], [4], [5], [7], [8] O-products-d115064cbf77

red-optical · Kitt Peak (1984, molecfit) · O I · Codex-graded lines · 1D NLTE (Bergemann) · [1], [3], [4], [5], [7], [8] O-products-0d45f4f81b47

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. source record

[2] — Amarsi et al. 2021, A&A 656 A113 DOI: 10.1051/0004-6361/202141384. source record

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. source record

[4] — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop (14-16 February 2006, UNLV, Las Vegas) source record

[5] — Wiese, W. L.; Fuhr, J. R.; Deters, T. M. (1996). Atomic Transition Probabilities of Carbon, Nitrogen, and Oxygen: A Critical Data Compilation. JPCRD Monograph 7. source record

[6] — Tachiev & Froese Fischer 2002, A&A 385, 716 DOI: 10.1051/0004-6361:20011816. source record

Model and grid sources

[7] — Amarsi, A. M.; Nissen, P. E.; Skuladottir, A. (2019). Carbon, oxygen, and iron abundances in disk and halo stars. A&A 630, A104 DOI: 10.1051/0004-6361/201936265. source record

[8] — Amarsi, Barklem, Asplund, Collet & Zatsarinny 2018, A&A 616, A89 DOI: 10.1051/0004-6361/201832770. source record

[9] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

Special diagnostics

[10] — Johansson, S.; Litzen, U.; Lundberg, H.; Zhang, Z. (2003). Experimental f-Value and Isotopic Structure for the Ni I Line Blended with [O I] at 6300 A. ApJ 584, L107 DOI: 10.1086/374037. source record

[11] — Storey, P. J.; Zeippen, C. J. (2000). Theoretical values for the [O III] 5007/4959 line-intensity ratio and homologous cases. MNRAS 312, 813 DOI: 10.1046/j.1365-8711.2000.03184.x. source record

[12] — Caffau, E.; Ludwig, H.-G.; Steffen, M.; Livingston, W.; Bonifacio, P.; Malherbe, J.-M.; Doerr, H.-P.; Schmidt, W. (2015). The photospheric solar oxygen project. III. Investigation of the centre-to-limb variation of the 630 nm [O I]–Ni I blend. A&A 579, A88. source record